Pearson

Use of AI in Design and Technology NEA

Agenda

Use of AI in D&T NEA

In this session we are going to look at:

- the concept of AI within the NEA
- AI misuse and spotting the signs
- using AI appropriately
- how AI can be used within the NEA
- how to declare responsible use.



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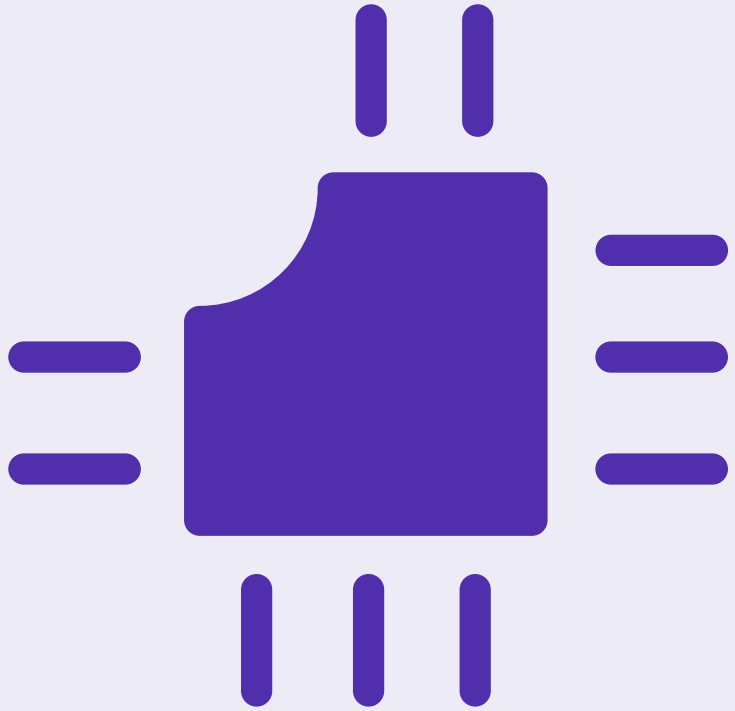
Introduction to AI

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What is AI?

AI stands for artificial intelligence and can be used to create the following based on instructions given to an AI tool:

- text
- images
- videos
- artwork
- CAD
- code.



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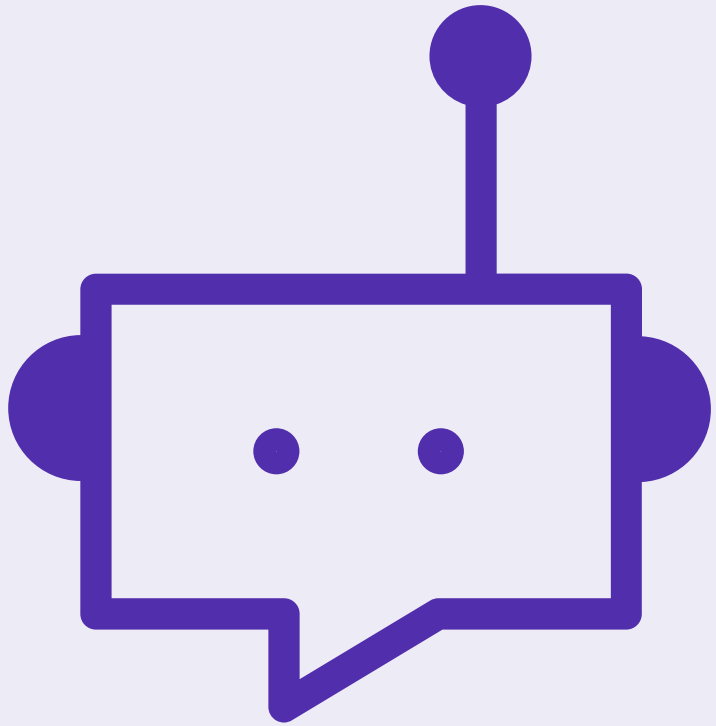
What is an AI Tool?

AI tools are applications you can use to create text, images, videos, artwork, code etc. These include, but not limited to:

- ChatGPT / Dali (Open AI)
- Claude (Anthropic)
- Gemini / Nano Banana (Google)
- Copilot (Microsoft) (uses ChatGPT).

These are not endorsed by Pearson and is for information purposes only.

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Why does AI matter?

Rapid student adoption

Students love a quick win!

Quick to adopt and keep ahead of the curve on new AI tools that come out.

Difficulty detecting misuse

AI is getting better and faster every month.

New products being updated which are superior to the last.

Getting harder to spot.

Exam board's expectations evolving

As AI changes and updates, what we accept changes too.

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AI Use & AI Misuse

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When can students use AI in their project?

As a **teacher**, you can use AI in class when teaching the course but remember teaching in the classroom is not an exam.

You can use AI to:

- plan lessons
- structure a slide deck
- make a presentation, audio files, videos, mind maps, quizzes, tests, revision materials etc.

When **students** use AI to produce work, they must acknowledge the AI tool, and the material they have produced using it.

If they **do not** acknowledge using AI, it will be considered 'AI misuse', otherwise known as cheating (malpractice).

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Referencing AI

When producing a piece of work, if you use the same wording as a published source, you must place quotation marks around the passage and state where it came from. This is known as referencing. Students must make sure that they give detailed references for everything in their work which is not in their own words, for example:

ChatGPT 3.5 (https://openai.com/blog/chatgpt/), 25/01/2025.

If students use an AI tool, they **cannot** get marks for what the AI tool has produced, as it is not their own work.

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What is AI Misuse?

- **Copying or paraphrasing sections** of AI-generated content so that the work is no longer the student's own.
- **Copying or paraphrasing whole responses** of AI-generated content.
- **Using AI to complete parts of the assessment** so that the work does not reflect the student's own work, analysis, evaluation or calculations.
- **Failing to acknowledge use of AI tools** when they have been used as a source of information.
- **Incomplete or poor acknowledgement of AI tools.**
- **Submitting work with intentionally incomplete** or misleading references or bibliographies.

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Spotting Signs of AI Misuse

- Unnaturally consistent sentence structure and length.
- Unusual vocabulary outside of usual range or the qualification syllabus / terminology.
- Sudden production of material at a pace which is at odds with the candidate's normal rate of production of work.
- Varying quality of work / varying levels of skillsets within one body of work.
- Comparison with previous work.
- References that cannot be found.
- Lack of direct quotes and references.
- Repetitive phrases or overuse of certain words.
- Less likely to include personal anecdotes, opinions or self reference.
- Lack of graphs and data tables where these would normally be expected.

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Why is it important to spot misuse of AI early?

- All exam boards (AOs) ensure that their moderators and examiners are appropriately trained in the identification of malpractice.
- Procedures are in place for reporting and investigating suspected malpractice.
- If AI misuse is suspected by a moderator or examiner, (or if it has been reported by a student, centre assessor or member of the public), full details of the allegation will usually be relayed to the centre (E6).
- Pearson will liaise with the Head of Centre regarding the next steps of the investigation and how appropriate evidence will be obtained.
- Pearson will then consider the case and, if necessary, impose a sanction in line with the sanctions given in the JCQ Suspected Malpractice: Policies and Procedures.

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How AI misuse is penalised

- The sanctions applied to a student committing plagiarism and making a false declaration of authenticity range from a warning regarding future conduct to disqualification and the student being barred from entering for one or more examinations for a set period of time.
- Pearson may also take action, which can include the imposition of sanctions, where centre staff are knowingly accepting, or failing to check, inauthentic work for qualification assessments.
- Moderators and examiners **will** treat AI misuse **exactly the same** as if they detected any other signs of plagiarism or copying.

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Investigation

Example

Rest of the portfolio

No Evidence of AI

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Product Comparison

			
Aesthetics	Bright, attracts attention	Big decals on the front	Bright, attracts attention from children
Cost	Cheap	Expensive, up to £1500	Not as expensive
Customer	Small businesses	Clothes collection charities	Family locations
Environment	Waterproof plastic, can go outside	Waterproof and weather resistant	Indoors
Size	Small, easy to carry	Large, suitable for clothes donation	Large, holds a lot of money
Safety	Smooth, no sharp edges	Heavy duty, large locks	Locks
Function	Collect coins	Collect clothes	Collect donations
Material	Plastic	Metal	Plastic and metal

Charity name: Outward bound

Bio:
At Outward Bound, young people take on adventures, overcome challenges and develop confidence and self-belief. These experiences are life changing.

Activities carried out:
Outdoor learning courses
Schools, colleges and youth groups
Early careers
Summer adventures

Projects they have worked on:
Generation green

Outdoor Gear: Tents, sleeping bags, backpacks, hiking boots, rain jackets, and other camping essentials.
Navigation Tools: Compasses, maps, GPS devices, and navigation aids to help participants navigate their surroundings.
First Aid Kits: Comprehensive first aid kits with supplies for treating minor injuries and emergencies during outdoor activities.
Outdoor Clothing: Durable and weather-resistant clothing suitable for various outdoor conditions.
Cooking Equipment: Lightweight camping stoves, cookware, and utensils for preparing meals during expeditions.
Safety Equipment: Helmets, harnesses, ropes, and other safety gear for activities like rock climbing or high ropes courses.
Educational Materials: Books, guides, and resources related to outdoor skills, leadership, and personal development.

Research into materials

Wood: Easy to work with and hard wearing
Acrylic: easy to cut and use for multiple purposes

Conclusion

Product Disassembly

Materials
Metal, use vinyl decals

Appearance
Bulky, solid colours

Function
Collect clothes donations for charities



Customer
Charities that help the homeless

Sustainability
Made using recycled metals, can be recycled or reused in a different location

Size
Very large, high capacity

Cost
Around £1000

Safety
Heavy duty locks, obstructions to prevent people climbing into it

Conclusion

Investigation

Example

Evidence of AI

Product Analysis







The Coin Vortex Charity Box is a unique and innovative product designed to encourage charitable giving in a fun and engaging way. This product utilizes a vortex-like mechanism that creates a mesmerizing effect when coins are dropped into it, attracting attention and encouraging people to donate One of the key strengths of the Coin Vortex Charity Box is its ability to capture the interest of individuals, making it an effective tool for fundraising events, retail stores, or any location where donations are sought. The visual appeal and interactive nature of the product can help increase donation rates and create a positive impact on fundraising efforts. Additionally, the Coin Vortex Charity Box is designed with convenience in mind. It features a secure and easy-to-use coin collection system, allowing for hassle-free retrieval of donations. This makes it suitable for both small and large-scale fundraising campaigns, as well as for personal use in homes or offices. Furthermore, the Coin Vortex Charity Box promotes transparency and accountability. It is equipped with a clear display that shows the amount of money collected, providing reassurance to donors that their contributions are being properly accounted for and utilized for charitable purposes. While the Coin Vortex Charity Box offers several advantages, it is important to consider potential limitations. The product may require occasional maintenance to ensure proper functionality, and its effectiveness may vary depending on the target audience and location. Additionally, it is crucial to have a well-planned marketing and communication strategy to maximize its impact and reach. In conclusion, the Coin Vortex Charity Box is an innovative and visually appealing product that can effectively engage individuals and encourage charitable giving. Its convenience, transparency, and interactive nature make it a valuable tool for fundraising campaigns and donation collection efforts.

The dog donation box is a unique and innovative product designed to facilitate the collection of donations for dogs in need. This product serves as an essential tool for animal shelters, rescue organizations, and other charitable initiatives aimed at supporting the well-being of dogs. One of the key features of the dog donation box is its eye-catching design. Typically, these boxes are shaped like dog houses or have adorable dog-themed graphics, making them visually appealing and capturing the attention of potential donors. This design element is essential as it encourages people to notice the box and feel compelled to contribute. In terms of functionality, the dog donation box is designed to be user-friendly and convenient. It is equipped with a lockable lid, ensuring the safety and security of the donations collected. Moreover, the box is usually made from durable materials such as acrylic or metal, which guarantees its longevity and ability to withstand regular use. The size of the dog donation box is another important factor to consider. It should be large enough to hold a significant amount of donations but not so big that it becomes cumbersome or difficult to manage. Ideally, the box should have a clear front panel or a slot for inserting cash or checks, allowing donors to easily contribute without any hassle. Furthermore, customization options are available for the dog donation box. Organizations can personalize the box with their logo, mission statement, or contact information, reinforcing their branding and making it easier for interested individuals to get in touch or learn more about the cause being supported. Overall, the dog donation box is an effective tool for fundraising and raising awareness for dogs in need. Its attractive design, user-friendly features, and customization options make it an ideal choice for organizations seeking to collect donations and support their efforts in providing care and assistance to dogs.

The metal clothes donation box is a practical and durable product designed to facilitate the collection of clothing donations for those in need. This product serves as an essential tool for charities, non-profit organizations, and community initiatives aimed at supporting individuals who require clothing assistance. One of the key features of the metal clothes donation box is its sturdy construction. Made from durable metal materials such as steel or aluminium, this box is built to withstand various weather conditions and ensure the safety of the donations collected. Its robustness guarantees that the box will last long-term, even when placed outdoors. The design of the metal clothes donation box is typically simple and functional. It often comes with a hinged or sliding door, allowing for easy access and retrieval of donated clothes. Some models may also feature a lockable mechanism to prevent theft or unauthorized access. This design element ensures the security of the donations and provides peace of mind for both donors and organizers. In terms of size, the metal clothes donation box is generally spacious enough to accommodate a significant amount of clothing donations. The box should be large enough to hold various sizes of clothing items, including jackets, pants, shirts, and accessories. Additionally, it should have proper ventilation to prevent odours and maintain the freshness of the donated clothes. Customization options are also available for the metal clothes donation box. Organizations can add their branding, logo, or contact information to the box, creating awareness about their cause and making it easier for individuals to connect or learn more about their initiatives. This personalization adds a professional touch and helps build trust and credibility with potential donors. Overall, the metal clothes donation box is an effective and practical tool for collecting clothing donations. Its durable construction, functional design, and customization options make it an ideal choice for organizations seeking to gather clothing items for those in need. By providing a secure and convenient donation method, this product plays a significant role in supporting communities and making a positive impact on the lives of individuals who require assistance.

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Investigation

Example

Evidence of AI

The dog donation box is a unique and innovative product designed to facilitate the collection of donations for dogs in need. This product serves as an essential tool for animal shelters, rescue organizations, and other charitable initiatives aimed at supporting the well-being of dogs. One of the key features of the dog donation box is its eye-catching design. Typically, these boxes are shaped like dog houses or have adorable dog-themed graphics, making them visually appealing and capturing the attention of potential donors. This design element is essential as it encourages people to notice the box and feel compelled to contribute. In terms of functionality, the dog donation box is designed to be user-friendly and convenient. It is equipped with a lockable lid, ensuring the safety and security of the donations collected. Moreover, the box is usually made from durable materials such as acrylic or metal, which guarantees its longevity and ability to withstand regular use. The size of the dog donation box is another important factor to consider. It should be large enough to hold a significant amount of donations but not

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Investigation

Example

Evidence of AI

Brief: idea 1/3

A VR user on a park bench, visibly disconnected from the physical environment, can be a tool for a public demonstration or social training. The performance can range from a public art piece to a form of therapy, using the user's isolation to highlight issues around modern social interaction.The contrast between virtual and reality: By showing a person isolated in a public space, the performance draws a contrast between the immersive virtual reality and the physical world passing them by. Passersby become part of the exhibit, reflecting on their own use of technology and the state of modern social connection.

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Authorship Violations

- ✕ Copying AI text as own work
- ✕ Submitting AI-generated designs without modification
- ✕ AI-written evaluation

Assessment Risks

- ✕ Malpractice concerns
- ✕ Loss of marks for authenticity
- ✕ Potential disqualification (extreme cases)

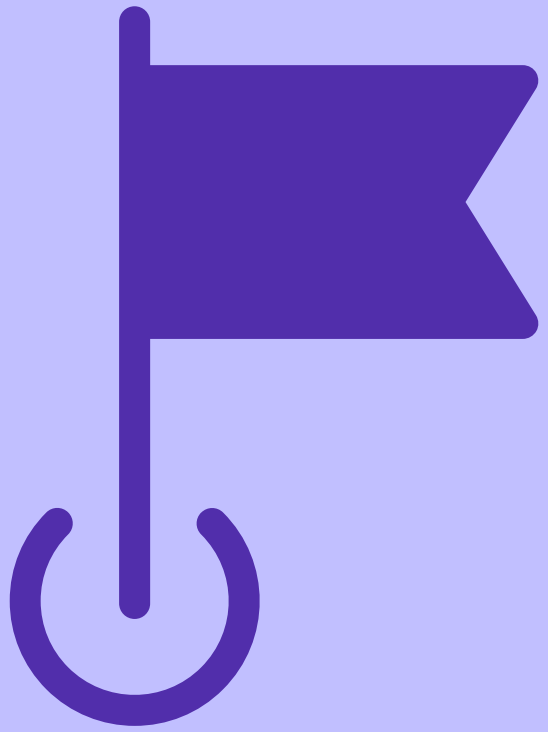
Design Integrity

- ✕ No evidence of iterative thinking
- ✕ Designs student cannot explain
- ✕ Overly “perfect” portfolios

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Red Flags to Spotting Misuse

- Sudden jump in quality
- Generic writing tone
- Lack of development



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Appropriate vs Inappropriate Use

AI is a **support tool.**

AI is not and should not be used as a **replacement for thinking.**

This is a **CREATIVE** subject and we should be building these skills and harnessing them, not taking the easy option.

It is **your responsibility** to observe misuse, through questioning and prompting and flagging where it has been used.

If a student cannot explain / justify / recreate it – **it is** likely it is inappropriate use.

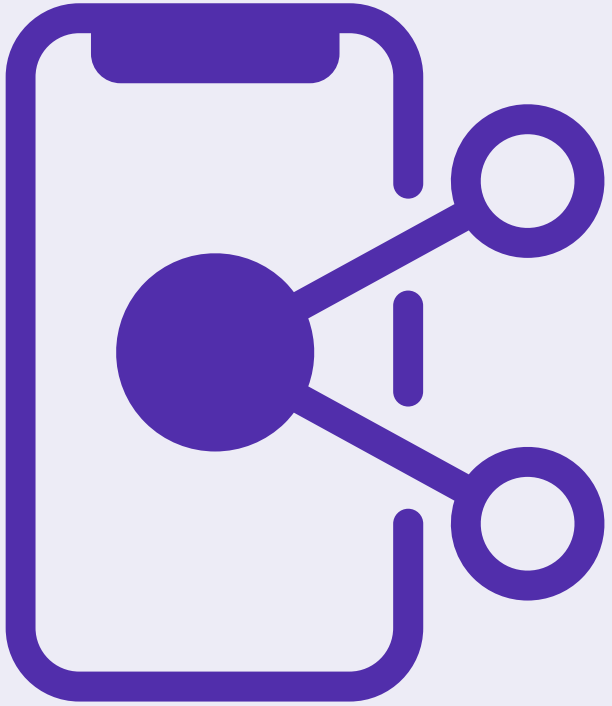
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Where does AI fit into the Design Process at a glance?

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Let’s reframe this…

- Should we be afraid of this or use it for good?
- AI can be used as a design tool, not a shortcut.
- It can be used to support:
  - Investigation
  - Design Ideas
  - Development
  - Communication
- Industry are using it, so why can’t students?
- We have to look at using it appropriately and signpost.



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AI across A Level NEA Assessment Criteria

A Level	Grid Title
Grid 1	Identification and investigation of a design possibility
Grid 2	Investigation of needs and research
Grid 3	Specification
Grid 4	Design ideas
Grid 5	Development of design ideas
Grid 6	Final design solution
Grid 7	Review of development and final idea
Grid 8	Communication of design ideas
Grid 9 Grid 10	Tools and equipment Quality and accuracy
Grid 11	Testing and evaluation

Weblink to AI Tool use by Grids (A Level)

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Let’s look at some examples in portfolios

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Caveat: At this early stage, we only have some examples of AI. Where we do not have examples, we will give examples of where and how AI can be used and will be accepted.

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A Level examples

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Grid 1 – Identification and investigation of a design possibility

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Ideas for how to use AI in Grid 1

Grid 1 should really start with a client/problem by looking at the needs and wants of the client, so AI should be a supplementary tool only. These are a few suggestions of how AI could be used.



Students could use AI to:

- find case studies of similar problems in architecture, scenarios, etc.
- transcribe an interview with a client/user group
- find the strengths and weakness in the transcription of an interview with a user group/client
- tell them more about the area they are looking into – then use this to discuss potential issues or further areas with the client/user group
- support with architecture by telling them more about the site, the surrounding amenities, any potential issues with the site, etc. Use this as part of their client conversation
- suggest questions they should be asking the client about their wants and needs.

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Different examples of starting points for conversations with clients – using AI

- Looking at issues that the elderly might have being out in the garden.
- Looking at the amenities on a site that you would like to regenerate.
- Getting the starting information about case studies similar to the area that you would like to improve.
- Looking at the scale/size of the area that you would like to improve.

CENTRAL NODE: Elderly People Enjoying the Garden

1. Physical & Environmental Risks

- Heat / cold sensitivity
- Dehydration
- Sunburn / UV exposure
- Slips and trips (uneven ground, wet surfaces)
- Overexertion (lifting, bending, digging)
- Allergies (pollen, mould, compost)

2. Cognitive & Safety Concerns

- Disorientation or wandering (especially dementia)
- Unsafe tool use
- Forgetting to hydrate or rest
- Difficulty recognising hazards

3. Biological & Wildlife Risks

- Insect stings (bees, wasps)
- Tick bites
- Toxic plants (touching or ingesting)
- Animal hazards (foxes, cats, rodents)

4. Mobility & Health Limitations

- Arthritis pain from repetitive movements
- Difficulty kneeling or standing
- Poor balance
- Reduced strength for lifting
- Vision problems affecting hazard awareness

5. Social & Emotional Factors

- Gardening alone → no help if something happens
- Frustration if tasks become difficult
- Loss of confidence
- Feeling isolated despite being outdoors



Comparable UK Park & Riverside Regeneration Case Studies

2. Alexandra & Ainsworth Estate Park Regeneration (London)

A community-led regeneration where residents secured Heritage Lottery funding to refurbish their park. Relevance to Windsor:

- heritage-sensitive landscape improvements
- community-driven design
- refurbishment rather than redevelopment

3. Hastings Observer Building & Riverside Public Realm Renewal

Part of a wider regeneration that revived derelict riverside and town-centre spaces into community-oriented uses. Relevance to Windsor:

- turning underused riverside land into active public space
- heritage-led regeneration

4. Skipton Cultural Hub & Public Realm Upgrade

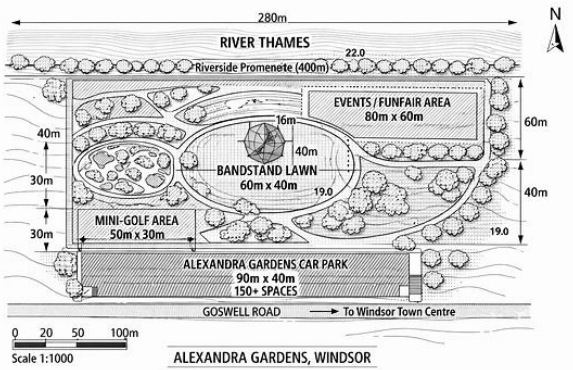
Award-winning redevelopment of a cultural hub with surrounding public-realm improvements. Relevance to Windsor:

- integrating cultural activity with public open space
- improving visitor experience and wayfinding

5. Oswestry Cross Street Revitalisation

Small-grants scheme that revitalised a historic town-centre street, improving public realm and reactivating vacant spaces. Relevance to Windsor:

- incremental improvements
- heritage-sensitive urban realm upgrades



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Grid 2 – Investigation of needs and research

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Ideas for how to use AI in Grid 2

Research requires a wholly client/user/candidate led justification/interpretation/analysis of the research. However, AI can be used as a source of research.

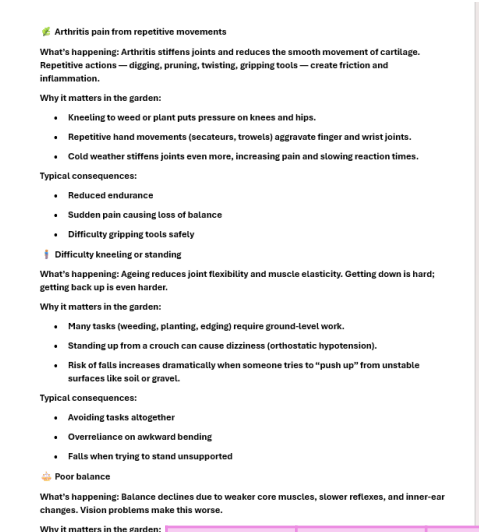
Students could use AI to:

- find previously mentioned case studies/designers/etc. who have encountered similar problems.
- analyse results from questionnaires and surveys
- research suitable properties of relevant materials
- assemble lots of data, such as anthropometrics
- generate ideas for questions to ask in surveys/client
- offer design inspiration or mood boards (candidate should then annotate this with their thoughts).

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Different examples of how AI might springboard research/client conversations

- Example of a previous review of the area and the issues found from a wide range of stakeholders.
- Data from anthropometrics collated.
- Suitable materials from a school list based on certain properties.
- More detailed look at issues with arthritis in the garden to use as a starting point for client discussion.



🌾 **Arthritis pain from repetitive movements**

What's happening: Arthritis stiffens joints and reduces the smooth movement of cartilage. Repetitive actions — digging, pruning, twisting, gripping tools — create friction and inflammation.

Why it matters in the garden:

- Kneeling to weed or plant puts pressure on knees and hips.
- Repetitive hand movements (secateurs, trowels) aggravate finger and wrist joints.
- Cold weather stiffens joints even more, increasing pain and slowing reaction times.

Typical consequences:

- Reduced endurance
- Sudden pain causing loss of balance
- Difficulty gripping tools safely

🚶 **Difficulty kneeling or standing**

What's happening: Ageing reduces joint flexibility and muscle elasticity. Getting down is hard; getting back up is even harder.

Why it matters in the garden:

- Many tasks (weeding, planting, edging) require ground-level work.
- Standing up from a crouch can cause dizziness (orthostatic hypotension).
- Risk of falls increases dramatically when someone tries to "push up" from unstable surfaces like soil or gravel.

Typical consequences:

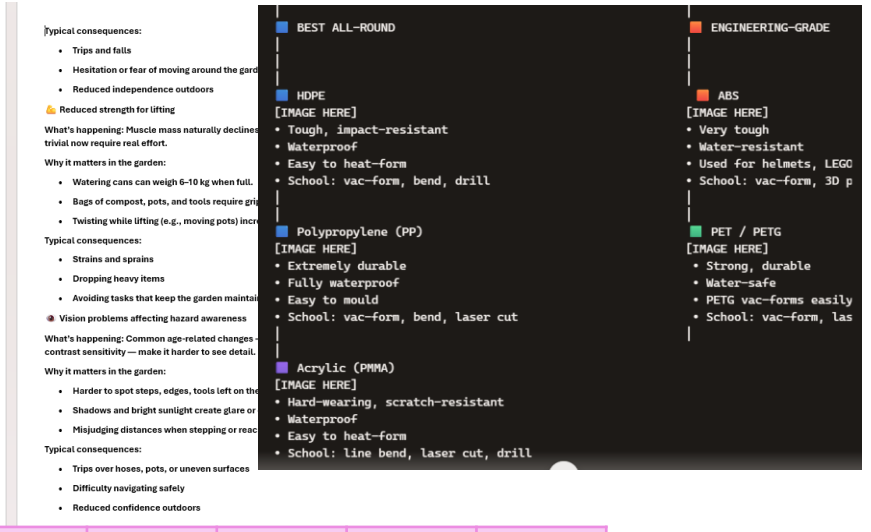
- Avoiding tasks altogether
- Overreliance on awkward bending
- Falls when trying to stand unsupported

⚖️ **Poor balance**

What's happening: Balance declines due to weaker core muscles, slower reflexes, and inner-ear changes. Vision problems make this worse.

Why it matters in the garden:

- Uneven ground, roots, [illegible]
- Wet grass or mossy pav[illegible]
- Carrying tools or water[illegible]



Typical consequences:

- Trips and falls
- Hesitation or fear of moving around the gard[illegible]
- Reduced independence outdoors

💪 **Reduced strength for lifting**

What's happening: Muscle mass naturally declines[illegible] trivial now require real effort.

Why it matters in the garden:

- Watering cans can weigh 6–10 kg when full.
- Bags of compost, pots, and tools require grig[illegible]
- Twisting while lifting (e.g., moving pots) incr[illegible]

Typical consequences:

- Strains and sprains
- Dropping heavy items
- Avoiding tasks that keep the garden maintai[illegible]

👁️ **Vision problems affecting hazard awareness**

What's happening: Common age-related changes [illegible] contrast sensitivity — make it harder to see detail.

Why it matters in the garden:

- Harder to spot steps, edges, tools left on the[illegible]
- Shadows and bright sunlight create glare or[illegible]
- Misjudging distances when stepping or reac[illegible]

Typical consequences:

- Trips over hoses, pots, or uneven surfaces
- Difficulty navigating safely
- Reduced confidence outdoors

🟦 BEST ALL-ROUND

🟦 HDPE
[IMAGE HERE]
• Tough, impact-resistant
• Waterproof
• Easy to heat-form
• School: vac-form, bend, drill

🟦 Polypropylene (PP)
[IMAGE HERE]
• Extremely durable
• Fully waterproof
• Easy to mould
• School: vac-form, bend, laser cut

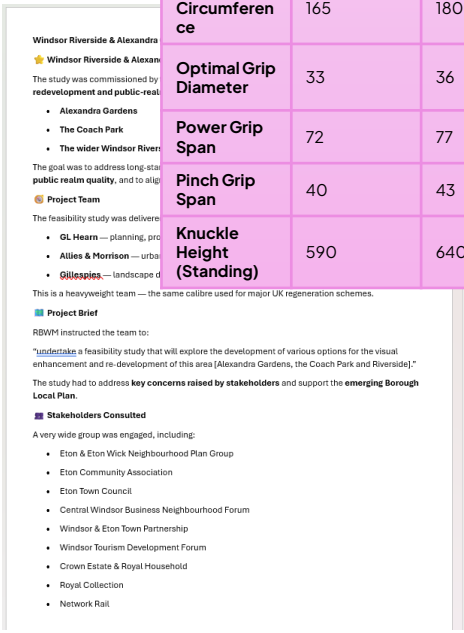
🟪 Acrylic (PMMA)
[IMAGE HERE]
• Hard-wearing, scratch-resistant
• Waterproof
• Easy to heat-form
• School: line bend, laser cut, drill

🟥 ENGINEERING-GRADE

🟥 ABS
[IMAGE HERE]
• Very tough
• Water-resistant
• Used for helmets, LEGO
• School: vac-form, 3D p

🟩 PET / PETG
[IMAGE HERE]
• Strong, durable
• Water-safe
• PETG vac-forms easily
• School: vac-form, las

Measure	Elderly Women 5th	Elderly Women 50th	Elderly Women 95th	Elderly Men 5th	Elderly Men 50th	Elderly Men 95th
Hand Length	160	172	185	175	188	205
Hand Breadth	72	78	86	82	90	98
Hand Circumference	165	180	200	185	205	225
Optimal Grip Diameter	33	36	40	37	41	45
Power Grip Span	72	77	83	79	85	92
Pinch Grip Span	40	43	46	44	47	51
Knuckle Height (Standing)	590	640	690	620	685	750



Windsor Riverside & Alexandra

⭐ **Windsor Riverside & Alexan[illegible]**

The study was commissioned by [illegible] **redevelopment and public-real[illegible]**

- **Alexandra Gardens**
- **The Coach Park**
- **The wider Windsor River[illegible]**

The goal was to address long-sta[illegible] **public realm quality**, and to alig[illegible]

🎯 **Project Team**

The feasibility study was delivere[illegible]

- **GL Hearn** — planning, pro[illegible]
- **Allies & Morrison** — urba[illegible]
- **Gillespies** — landscape d[illegible]

This is a heavyweight team — the same calibre used for major UK regeneration schemes.

🗺️ **Project Brief**

RBWM instructed the team to:

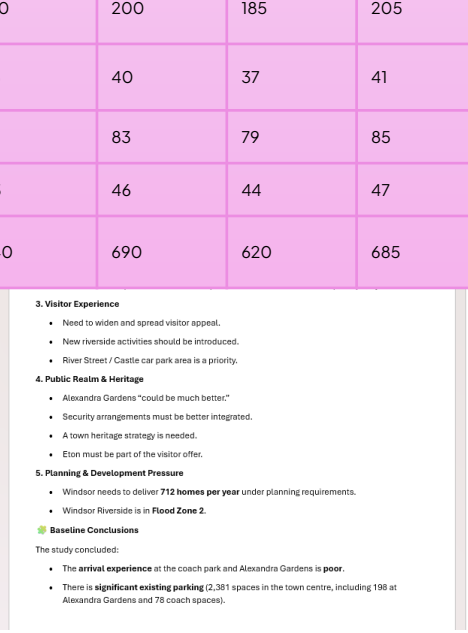
"undertake a feasibility study that will explore the development of various options for the visual enhancement and re-development of this area [Alexandra Gardens, the Coach Park and Riverside]."

The study had to address **key concerns raised by stakeholders** and support the **emerging Borough Local Plan**.

👥 **Stakeholders Consulted**

A very wide group was engaged, including:

- Eton & Eton Wick Neighbourhood Plan Group
- Eton Community Association
- Eton Town Council
- Central Windsor Business Neighbourhood Forum
- Windsor & Eton Town Partnership
- Windsor Tourism Development Forum
- Crown Estate & Royal Household
- Royal Collection
- Network Rail



3. Visitor Experience

- Need to widen and spread visitor appeal.
- New riverside activities should be introduced.
- River Street / Castle car park area is a priority.

4. Public Realm & Heritage

- Alexandra Gardens "could be much better."
- Security arrangements must be better integrated.
- A town heritage strategy is needed.
- Eton must be part of the visitor offer.

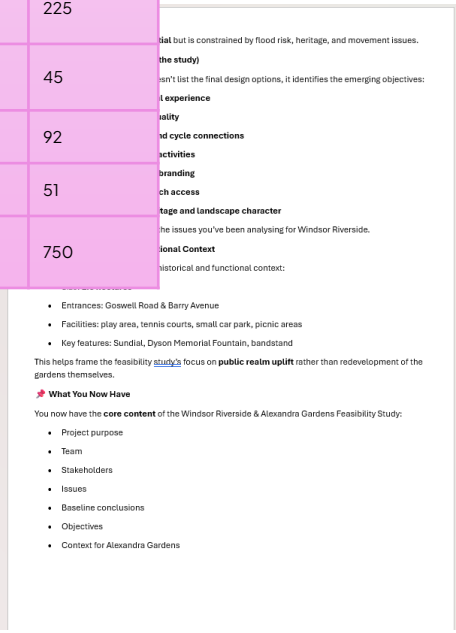
5. Planning & Development Pressure

- Windsor needs to deliver **712 homes per year** under planning requirements.
- Windsor Riverside is in **Flood Zone 2**.

🧩 **Baseline Conclusions**

The study concluded:

- The **arrival experience** at the coach park and Alexandra Gardens is **poor**.
- There is **significant existing parking** (2,381 spaces in the town centre, including 198 at Alexandra Gardens and 78 coach spaces).



dal but is constrained by flood risk, heritage, and movement issues.

the study)

sn't list the final design options, it identifies the emerging objectives:

t experience

uality

d cycle connections

ctivities

branding

ch access

tage and landscape character

he issues you've been analysing for Windsor Riverside.

ional Context

historical and functional context:

- Entrances: Goswell Road & Barry Avenue
- Facilities: play area, tennis courts, small car park, picnic areas
- Key features: Sundial, Dyson Memorial Fountain, bandstand

This helps frame the feasibility study's focus on **public realm uplift** rather than redevelopment of the gardens themselves.

📌 **What You Now Have**

You now have the **core content** of the Windsor Riverside & Alexandra Gardens Feasibility Study:

- Project purpose
- Team
- Stakeholders
- Issues
- Baseline conclusions
- Objectives
- Context for Alexandra Gardens

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Grid 3 – Specification

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Ideas for how to use AI in Grid 3

It is imperative that this section is scrutinised and justified by the candidate. However, AI can be used as a starting point for the candidate to then fully develop their own specification that is fully measurable and justified.



Students could use AI to:

- sort through the research and compile it into themes such as user requirements, costs, ergonomics, environment, performance requirements, etc.
- suggest some specification points for the candidate to consider as a starting point before developing them further to ensure that they are fully measurable and justified
- link specification points to research
- take free prose specification and turn it into bullet points for ease of testing in the design section.

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Grid 4 – Design ideas

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Ideas for how to use AI in Grid 4

Candidates must be able to communicate and annotate their designs sufficiently, in consultation with the client, to show their own understanding of tools, materials and manufacturing processes – as well as communication skills. However, AI can be used to supplement this.



Students could use AI:

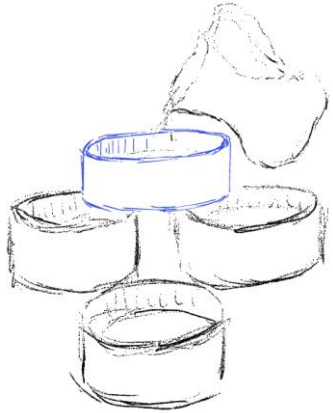
- as a design inspiration
- as a rendering tool
- as a starting point for solutions
- to research aspects relevant to the design.

AI cannot be used as the sole instrument to draw/render and annotate the ideas. Candidate input is essential and this must be shown.

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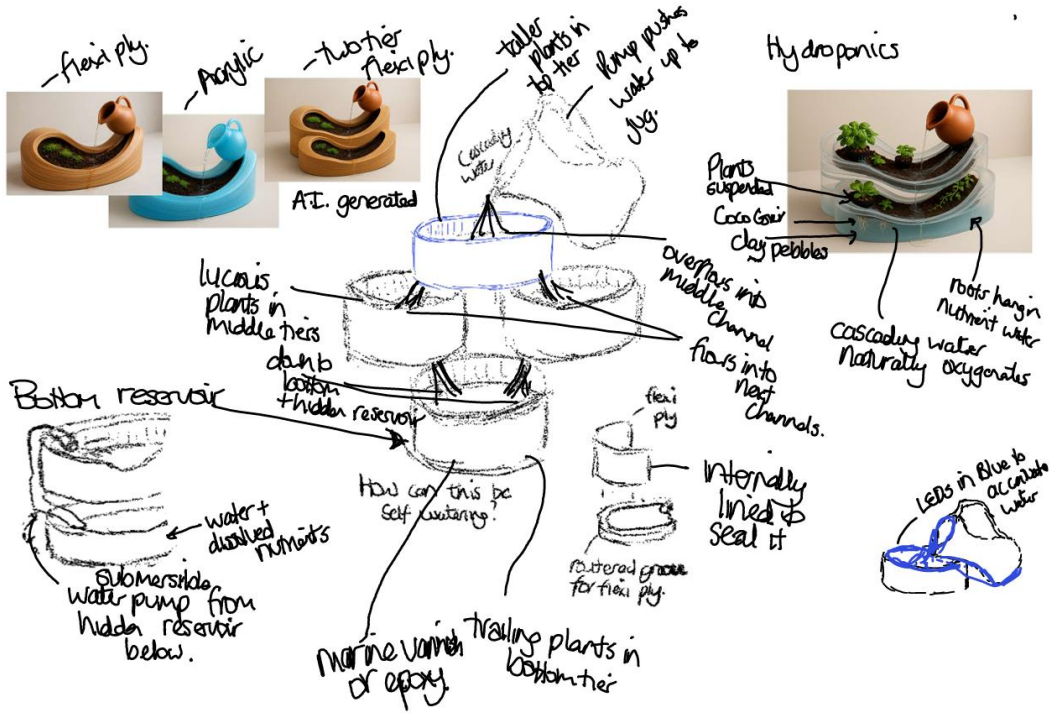
Different examples of how AI can be used as a design strategy

- Candidate used a sketch as a starting point.
- They then used this to render in different materials.
- The candidate then explained how the hydroponics should be added into the rendering.
- They then researched hydroponics and annotated the drawing to include these in with construction details.









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Grid 5 – Development of design ideas

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Ideas for how to use AI in Grid 5

Development must test aspects of the design in consultation with the client in order to springboard the idea forward. AI can be used to test aspects of this.



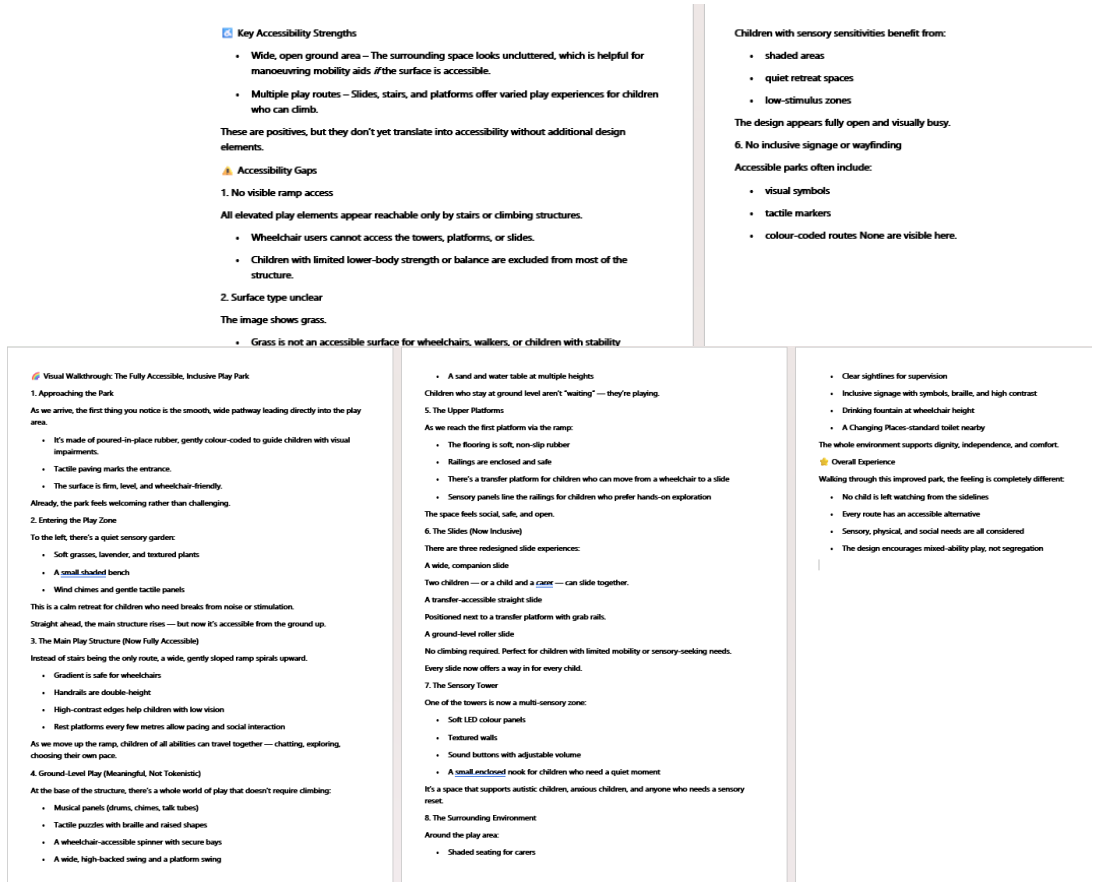
Students could use:

- AI to simulate testing of different design options for things such as:
  - structural performance and strength
  - weight reduction
  - material efficiency
  - environmental impact
  - walk through simulations.
- AI tools to develop and critique designs, to help products to meet diverse needs and consider accessibility.

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Different examples of how AI might be used in developing the idea

- The candidate has put the developed drawing through AI to evaluate the accessibility of a play park.
- They then used AI to identify the accessibility needs to address, helping to guide the early stages of design development and support discussions with the client on how the design could be refined.
- Another candidate has put their drawings into AI and asked it to render them in different colours to later liaise with the client.



🧭 Key Accessibility Strengths

- Wide, open ground area – The surrounding space looks uncluttered, which is helpful for manoeuvring mobility aids if the surface is accessible.
- Multiple play routes – Slides, stairs, and platforms offer varied play experiences for children who can climb.

These are positives, but they don't yet translate into accessibility without additional design elements.

⚠️ Accessibility Gaps

1. No visible ramp access

All elevated play elements appear reachable only by stairs or climbing structures.

- Wheelchair users cannot access the towers, platforms, or slides.
- Children with limited lower-body strength or balance are excluded from most of the structure.

2. Surface type unclear

The image shows grass.

- Grass is not an accessible surface for wheelchairs, walkers, or children with stability

Children with sensory sensitivities benefit from:

- shaded areas
- quiet retreat spaces
- low-stimulus zones

The design appears fully open and visually busy.

6. No inclusive signage or wayfinding

Accessible parks often include:

- visual symbols
- tactile markers
- colour-coded routes None are visible here.

🌈 Visual Walkthrough: The Fully Accessible, Inclusive Play Park

1. Approaching the Park

As we arrive, the first thing you notice is the smooth, wide pathway leading directly into the play area.

- It's made of poured-in-place rubber, gently colour-coded to guide children with visual impairments.
- Tactile paving marks the entrance.
- The surface is firm, level, and wheelchair-friendly.

Already, the park feels welcoming rather than challenging.

2. Entering the Play Zone

To the left, there's a quiet sensory garden:

- Soft grasses, lavender, and textured plants
- A small shaded bench
- Wind chimes and gentle tactile panels

This is a calm retreat for children who need breaks from noise or stimulation.

Straight ahead, the main structure rises — but now it's accessible from the ground up.

3. The Main Play Structure (Now Fully Accessible)

Instead of stairs being the only route, a wide, gently sloped ramp spirals upward.

- Gradient is safe for wheelchairs
- Handrails are double-height
- High-contrast edges help children with low vision
- Rest platforms every few metres allow pacing and social interaction

As we move up the ramp, children of all abilities can travel together — chatting, exploring, choosing their own pace.

4. Ground-Level Play (Meaningful, Not Tokenistic)

At the base of the structure, there's a whole world of play that doesn't require climbing:

- Musical panels (drums, chimes, talk tubes)
- Tactile puzzles with braille and raised shapes
- A wheelchair-accessible spinner with secure bays
- A wide, high-backed swing and a platform swing
- A sand and water table at multiple heights

Children who stay at ground level aren't "waiting" — they're playing.

5. The Upper Platforms

As we reach the first platform via the ramp:

- The flooring is soft, non-slip rubber
- Railings are enclosed and safe
- There's a transfer platform for children who can move from a wheelchair to a slide
- Sensory panels line the railings for children who prefer hands-on exploration

The space feels social, safe, and open.

6. The Slides (Now Inclusive)

There are three redesigned slide experiences:

A wide, companion slide

Two children — or a child and a carer — can slide together.

A transfer-accessible straight slide

Positioned next to a transfer platform with grab rails.

A ground-level roller slide

No climbing required. Perfect for children with limited mobility or sensory-seeking needs.

Every slide now offers a way in for every child.

7. The Sensory Tower

One of the towers is now a multi-sensory zone:

- Soft LED colour panels
- Textured walls
- Sound buttons with adjustable volume
- A small enclosed nook for children who need a quiet moment

It's a space that supports autistic children, anxious children, and anyone who needs a sensory reset.

8. The Surrounding Environment

Around the play area:

- Shaded seating for carers
- Clear sightlines for supervision
- Inclusive signage with symbols, braille, and high contrast
- Drinking fountain at wheelchair height
- A Changing Places-standard toilet nearby

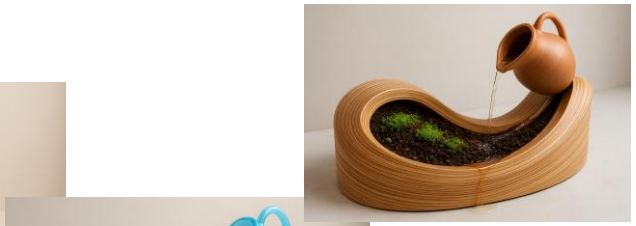
The whole environment supports dignity, independence, and comfort.

⭐ Overall Experience

Walking through this improved park, the feeling is completely different:

- No child is left watching from the sidelines
- Every route has an accessible alternative
- Sensory, physical, and social needs are all considered
- The design encourages mixed-ability play, not segregation







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Grid 6 – Final design solution

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Ideas for how to use AI in Grid 6

The candidate needs to be able to communicate their final design to a third party independently. However, AI can be used to support this section in the following ways.



Students could use AI:

- for a relevant manufacturing specification, focused on appropriate operations available to the candidate, within school workshop environment
- to take their operations and put them into a flow diagram
- to complete a complete risk assessment of the make
- to add quality control check points in
- to find information about brought in components.

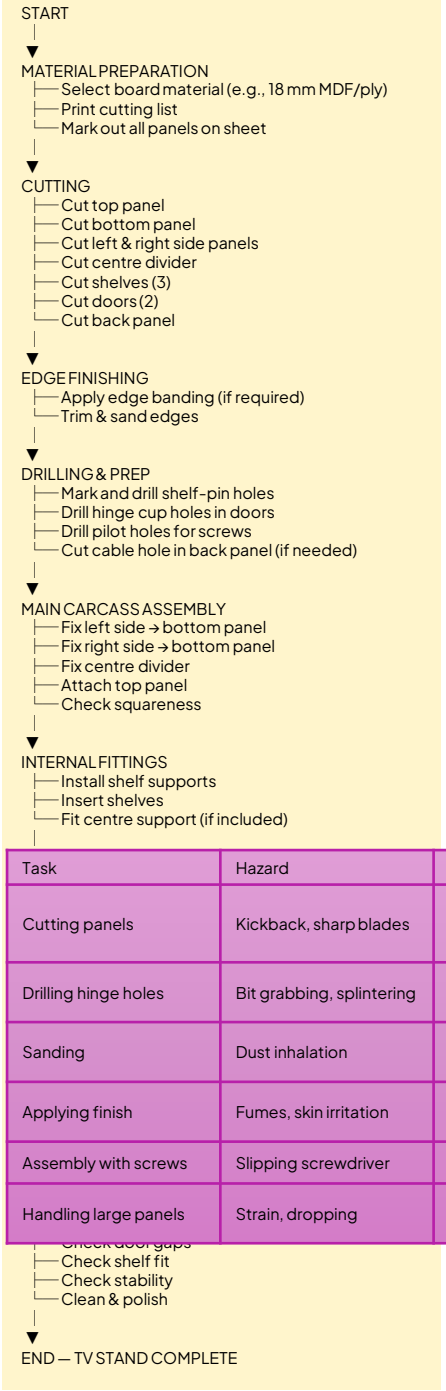
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Different examples of how AI may be used to support third party manufacture

The candidate has created a plan and put this into AI to create a flow chart.

They have used their plans to generate a cutting list.

They have asked AI to pull the safety factors and quality control from their plan to create checks along the way in manufacture.



Part	Qty	Dimensions (mm)	Notes
Top panel	1	1400 × 360 × 18	
Bottom panel	1	1400 × 360 × 18	
Side panels	2	360 × 405 × 18	Left & right
Centre divider	1	360 × 369 × 18	
Shelves	3	~450 × 360 × 18	L / R / Centre
Back panel	1	1400 × 360 × 6	
Doors	2	~450 × 330 × 18	
Centre support	1	360 × ~80 × 18	
Plinth/feet strips	3	see above	optional

Stage	Checkpoint	Tolerance	Method of Checking	Sign-off
Cutting	Panel dimensions	±1 mm	Steel rule, square	✓
Drilling	Hinge cup position	±0.5 mm	Jig alignment, test fit	✓
Assembly	Carcass squareness	±2 mm diagonally	Measure diagonals	✓
Shelves	Shelf level	0 mm deviation	Spirit level	✓
Doors	Gap consistency	2–3 mm	Visual check, feeler gauge	✓
			ection, nt	✓

Task	Hazard	Risk Level	Control Measures	PPE
Cutting panels	Kickback, sharp blades	High	Use push sticks, set blade guard, maintain correct stance	Safety glasses, apron
Drilling hinge holes	Bit grabbing, splintering	Medium	Clamp work securely, drill at correct speed	Safety glasses
Sanding	Dust inhalation	Medium	Use extraction, sand in ventilated area	Dust mask
Applying finish	Fumes, skin irritation	Medium	Use water-based finishes, ventilate area	Gloves, mask
Assembly with screws	Slipping screwdriver	Low	Pre-drill holes, use correct bit	None required
Handling large panels	Strain, dropping	Medium	Lift with assistance, support panels	Safety shoes

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Grid 7 – Review of development and final idea

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Ideas for how to use AI in Grid 7

As this grid shows the analysis and evaluative skills of the candidate, there is limited use for AI in this section. However, the use of AI in grids 4-6 may well catapult the design thinking and offer some balanced conversations with the client.



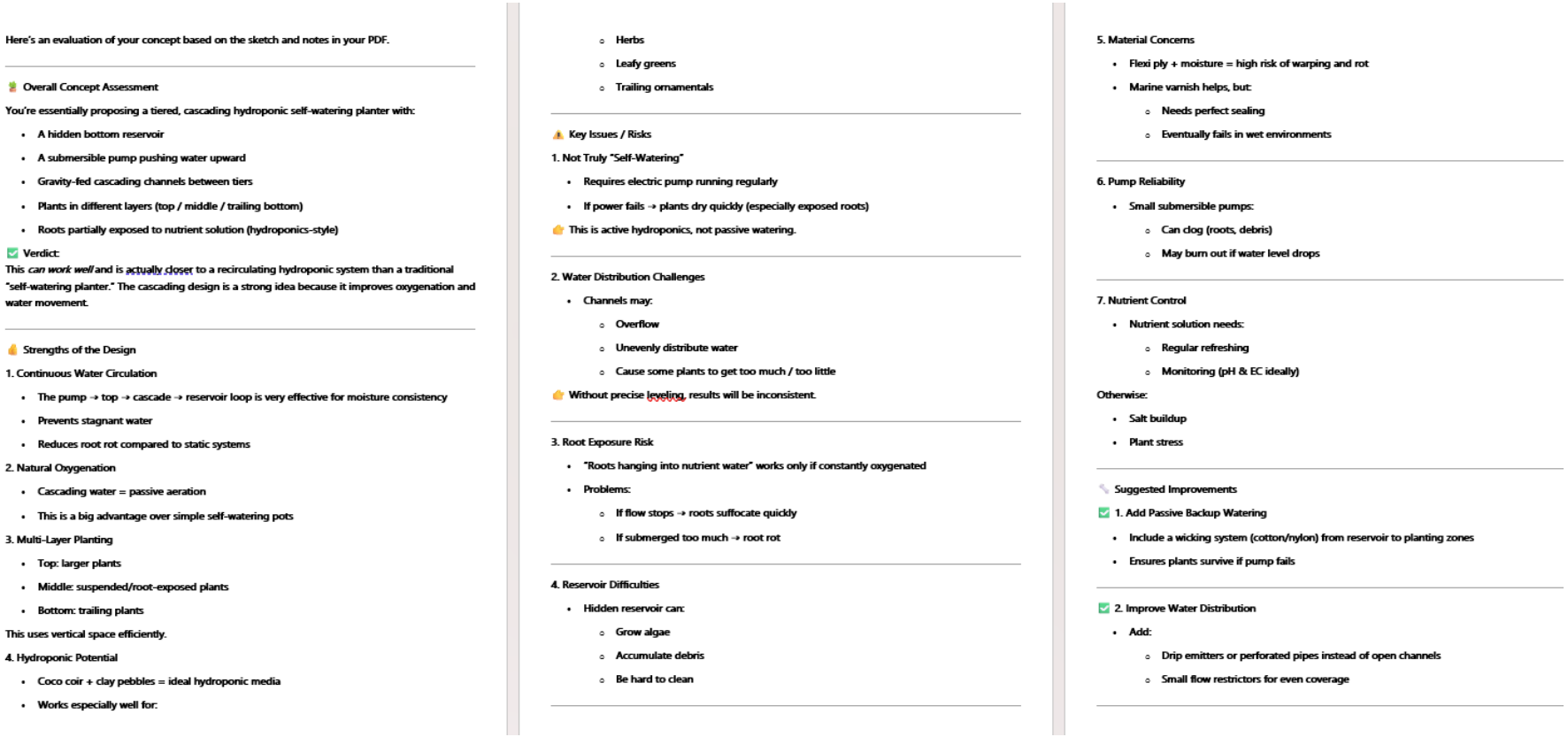
Students could use AI to:

- offer feedback on development with the client/user to test
- offer balanced discussions in this section.

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Different examples of how AI used to “test” aspects of the design in order to support analysis and evaluation

AI has been used to evaluate different designs and offer thoughts for further development and discussion with the client.



Here's an evaluation of your concept based on the sketch and notes in your PDF.

🪴 Overall Concept Assessment

You're essentially proposing a tiered, cascading hydroponic self-watering planter with:

- A hidden bottom reservoir
- A submersible pump pushing water upward
- Gravity-fed cascading channels between tiers
- Plants in different layers (top / middle / trailing bottom)
- Roots partially exposed to nutrient solution (hydroponics-style)

✅ Verdict:
This *can work well* and is actually closer to a recirculating hydroponic system than a traditional "self-watering planter." The cascading design is a strong idea because it improves oxygenation and water movement.

👍 Strengths of the Design

1. Continuous Water Circulation
   - The pump → top → cascade → reservoir loop is very effective for moisture consistency
   - Prevents stagnant water
   - Reduces root rot compared to static systems
2. Natural Oxygenation
   - Cascading water = passive aeration
   - This is a big advantage over simple self-watering pots
3. Multi-Layer Planting
   - Top: larger plants
   - Middle: suspended/root-exposed plants
   - Bottom: trailing plants

This uses vertical space efficiently.

4. Hydroponic Potential
   - Coco coir + clay pebbles = ideal hydroponic media
   - Works especially well for:
     - Herbs
     - Leafy greens
     - Trailing ornamentals

⚠️ Key Issues / Risks

1. Not Truly "Self-Watering"
   - Requires electric pump running regularly
   - If power fails → plants dry quickly (especially exposed roots)

👉 This is active hydroponics, not passive watering.

---

2. Water Distribution Challenges
   - Channels may:
     - Overflow
     - Unevenly distribute water
     - Cause some plants to get too much / too little

👉 Without precise leveling, results will be inconsistent.

---

3. Root Exposure Risk
   - "Roots hanging into nutrient water" works only if constantly oxygenated
   - Problems:
     - If flow stops → roots suffocate quickly
     - If submerged too much → root rot

---

4. Reservoir Difficulties
   - Hidden reservoir can:
     - Grow algae
     - Accumulate debris
     - Be hard to clean

---

5. Material Concerns
   - Flexi ply + moisture = high risk of warping and rot
   - Marine varnish helps, but:
     - Needs perfect sealing
     - Eventually fails in wet environments

---

6. Pump Reliability
   - Small submersible pumps:
     - Can clog (roots, debris)
     - May burn out if water level drops

---

7. Nutrient Control
   - Nutrient solution needs:
     - Regular refreshing
     - Monitoring (pH & EC ideally)

Otherwise:

- Salt buildup
- Plant stress

🦴 Suggested Improvements

✅ 1. Add Passive Backup Watering
   - Include a wicking system (cotton/nylon) from reservoir to planting zones
   - Ensures plants survive if pump fails

✅ 2. Improve Water Distribution
   - Add:
     - Drip emitters or perforated pipes instead of open channels
     - Small flow restrictors for even coverage

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Grid 8 – Communication of design ideas

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Ideas for how to use AI in Grid 8

This has already been covered in the other sections. In summary, AI could be used to enhance communication in the following ways.

Students could use AI:

- as a rendering tool
- as a design strategy
- to generate schedule of work and QC/risk assessments.

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Grid 9 – Tools and equipment & Grid 10 – Quality and accuracy

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Ideas for how to use AI in Grid 9 and 10

This section requires the candidate to undertake their own manufacture. As such any CAD/CAM/AI that is used must just serve as one part of the manufacture. Examples of this are as follows.



- The candidate has used AI but adjusted the file to create parts that connect and independently develop how that affects quality and accuracy as a whole.
- The candidate may gain credit for the skills and processes involved in their own use of a 3D printer to produce the part using the AI tool generated STL.
- The candidate should create their own STL file using CAD software in order to gain credit for Grids 9 and 10. The candidate could critique the quality, accuracy and general performance of the AI tool generated STL or 3D printed product in order to inform their own independent use of CAD/CAM for the final prototype.

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Grid 11 – Testing and evaluation

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Ideas for how to use AI in Grid 11

Although AI can be used to test aspects of the prototype, the evaluation part of this section must be wholly completed by the candidate. The candidate pertinently used AI to test aspects of their product as well as their own in use and in situ testing; testing with the client/user group and testing against the specification. They have also undertaken their own independent evaluation



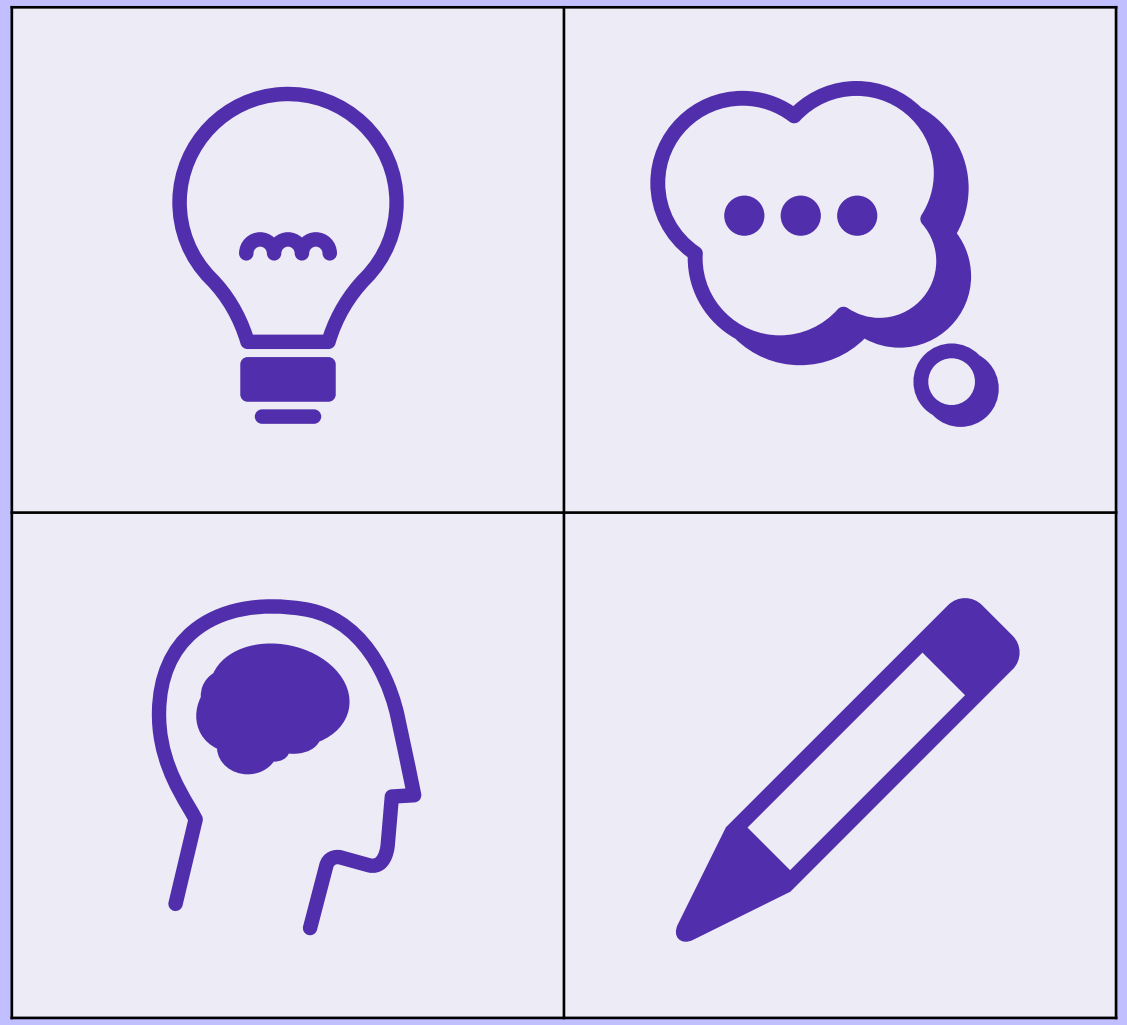
Students could use AI:

- to simulate testing different design options for things such as strength and structural performance; weight reduction; material efficiency; environmental impact; walk through simulations from drawings
- to test virtual product testing
- to improve accuracy of predictions to offer faster iteration cycles
- for lifecycle analysis
- to suggest modifications that the prototype would need for commercial manufacture.

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What skills do we want to see?

- Personal response to context
- Identify real users
- Generate original ideas
- Iterative design process
- Make design decisions
- Evaluate based on their product



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Declarations and Authentication

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Supporting AI Use not AI Misuse

Declare

- Students must sign their candidate authentication form to inform the teacher and the exam board that they are saying that all the work they are submitting as part of their NEA is their own.
- If they have used an AI tool, they must declare this, before they sign the declaration.

Acknowledge

- Reference the AI tool they have used.
- Give the date of when the AI tool generated the content.
- Give details of how they used it.
- Save screenshots of the prompt (what they asked / instructed the AI tool to do and what answer the AI gave them) and include this with the work submitted.
- Annotate how AI helped.

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Reference example

Reference page

Here are all the links for images I used from the internet in my coursework. I have no ownership over any of the images on this slide. All other photos in my coursework are take by me. Furthermore I used AI tools such as ChatGpt and Google Gemini to assist me in my Coursework and building the product. I only used them to research, gain information, and to use technical terms. I did not copy paste directly, rather wrote everything in my own words

PICTURES

MOODBOARD: PAGE 4

Puzzle: https://www.ebay.co.uk/itm/403149309324
Whiteboard: https://viz-pro.com/product/viz-pro-childs-drawing-writing-whiteboard/
Fidget spinner: https://thesensorysloth.com/products/fidget-spinner-fidget-toy?srsltid=AfmBOoqjclzRed8HIOXGjBPJday5oKlunnPCry3a-3-ilLSDjQBN5k94
Pop it: https://among-us.fr/pop-it/multicolore
Sensory toys: https://store.stageslearning.com/products/sensory-builder-sensory-kit
Wooden blocks: https://www.amazon.sg/Building-Stacking-Carrying-Montessori-Learning/dp/B0D7KMGKF6
Flash cards: https://www.amazon.in/STEMTICS-Knowledge-Learning-Educational-Flashcards/dp/B0FQ2MPHTF
Abacus: https://mybtoys.com/shop/two-ty-fruity-2/

ANTHROPOMETRICS AND ERGONOMICS: PAGE 10

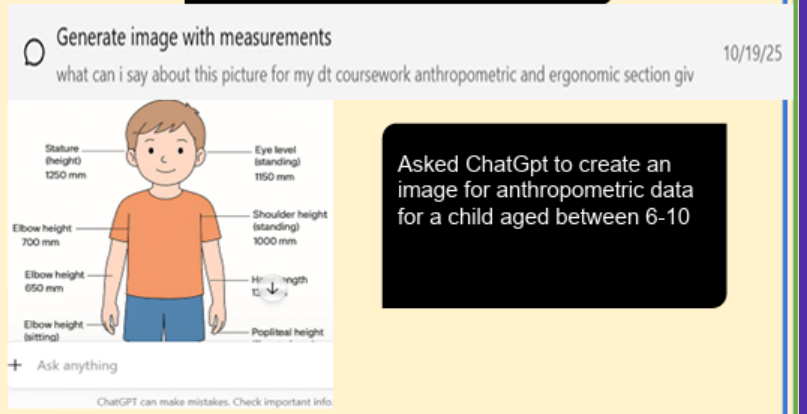
Posture: https://milestoneprc.com/blog/posture-in-children-causes-effects-correction/
Table and chair dimensions: https://www.pinterest.com/pin/336503403409125498/
Child standing and sitting dimensions: https://www.pinterest.com/pin/336503403409125498/
Hand dimensions: https://www.researchgate.net/figure/Finfer-length-Distance-between-proximal-flexion-crease-and-the-tip-finger_fig2_377553365
Sustainable products: https://collectlo.com/indrashis-609140/the-urgent-crisis-of-deforestation
Product life cycle analysis: https://ecochain.com/blog/life-cycle-assessment-lca-guide/

MATERIAL RESEARCH: PAGE 23

Acrylic sheets: https://jycaplastics.com.au/product/transparent-colour-acrylic/
Cans of paints: https://hendrickpainting.com/blog/how-to-choose-the-best-paint-finish-for-interior-walls-in-your-home/
Protector spray can: https://www.tarrago.com/product/trekking-protector-spray/

ADDITIONAL TOOLS:
Canva (To create all the tables on page 5,12,13,19,22 and 32): https://www.canva.com/
Google forms(To create the questionnaire for parents,restaurant staff and kids on page 6,7,8,9 : https://docs.google.com/forms/u/0/?tgif=d

Chat Gpt and Gemini



Asked ChatGpt to create an image for anthropometric data for a child aged between 6-10

Asked Gemini for information about joints

Woodworking Joints: Dado, Groove, Rabbet — 5 Jan
This is my **dt** initial research design so 1/4. This is a toy storage box which ...

Asked Gemini if i should do technical design or not

Hand-Drawn Technical Drawings Advantage — 27 Feb
Going the old-school route for your Design and Technology (**DT**) coursew...

Asked Gemini the name of some of the tools in my DT workshop and asked it to identify them

Common Design and Technology Tools — 13 May
Since you haven't specified which tool you are looking at, here are the mo...

Asked Gemini about the uses and qualities of materials like wood, styrofoam and acrylic

XPS Foam: Identification and Uses — 27 Feb
In a Design and Technology (**DT**) workshop, when working with wood (like ...

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A Level D&T Authentication

Pearson Edexcel

Pearson Edexcel Level 3 Advanced GCE in Design and Technology (Product Design) (9DT0)

Component 2

Candidate declaration

I certify that the work submitted for this assessment is my own. I have clearly referenced any sources and any AI tools used in the work. I understand that false declaration is a form of malpractice.

Candidate signed:		Date:	

Assessor name:			
Assessor signed:		Date:	

Candidate declaration

I certify that the work submitted for this assessment is my own. I have clearly referenced any sources and any AI tools used in the work. I understand that false declaration is a form of malpractice.

Candidate signed:		Date:	

Additional candidate declaration

By signing this additional declaration, you agree to your work being used to support professional development, online support and training of both centre-assessors and Pearson moderators. If you have any concerns, please email: dt.eng@pearson.com

Candidate signed:		Date:	

Supporting AI Use not AI Misuse

Teaching AI Literacy

- Show good vs bad prompts
- Model ethical use

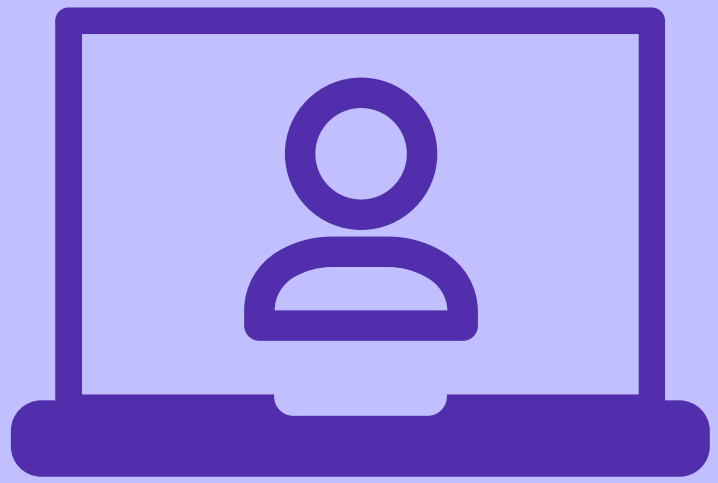
Controlled Use

- Use AI in lessons with guidance
- Not in final NEA write-up stages

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Key Takeaways

- AI is here to stay
- It can enhance creativity
- It must not replace thinking
- Consistency is critical



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JCQ Guidance & Documents

Teachers are recommended to read the Joint Council for Qualifications (JCQ) guidance documents relating to coursework, malpractice, and AI, which can be found using the following links.

AI Use in Assessments: Protecting the Integrity of Qualifications

Information for candidates Coursework assessments

Suspected Malpractice policies and procedures

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Summary

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Summary

In this session we looked at the following objectives:

- understand what AI is (in education context)
- explore opportunities in D&T NEA
- clarify appropriate vs inappropriate NEA use
- develop consistent department practice.

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